

Identifying the Origin of the Rate–Distortion Ceiling in Scalar-Quantized Neural Audio Codecs

Project Exposé

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Goal

This project determines, through controlled experimentation, whether the rate–distortion ceiling of scalar-quantized, non-adversarially-trained neural audio codecs originates in the resolution of the quantizer or in the latent entropy shaped by the training objective — clarifying whether the quality gap to adversarially-trained, vector-quantized systems such as EnCodec is fundamental to scalar quantization or a consequence of addressable design choices.

Problem

Neural audio codecs have replaced hand-engineered pipelines for extremely low-bitrate speech compression. Real-time use adds a causality constraint: no access to future samples. A central design choice is how the encoder's continuous latent is discretized for transmission. Scalar quantization (SQ) — rounding each latent dimension independently, paired with a conventional entropy coder — is simple and cheap but consistently underperforms more complex systems at matched bitrate. The literature has not isolated why: insufficient resolution, the absence of cross-dimension code assignment, the absence of adversarial training, or some combination remain equally plausible explanations. This project isolates the cause, to determine whether the gap can be closed without adopting the added complexity of adversarial, vector-quantized systems.

State of the Art

SoundStream (Zeghidour et al., 2021) and EnCodec (Défossez et al., 2023) achieve strong quality below 6 kbps using Residual Vector Quantization — an extension of the codebook-based quantization introduced by van den Oord et al. (2017) — combined with adversarial training. Brendel et al. (2024) show constant-bitrate scalar quantization is viable for real-time coding, but do not characterize what limits its quality. Concurrent work by Xu et al. (2026) pairs scalar quantization with a learned entropy model and reports substantial rate–distortion gains, indicating entropy-coding quality matters — but without isolating whether a fixed-coder SQ system's ceiling comes from the coder or from the latent itself. No existing work varies the training objective of an SQ codec under a fixed entropy coder to attribute the ceiling to one cause specifically.

Approach

This question is investigated empirically using a causal, streaming-capable codec: causal convolutions and a causal Transformer produce a 32-dimensional latent at 100 Hz from 16 kHz input, compressed via uniform 3-bit scalar quantization and a conventional, non-learned entropy coder (zlib). zlib is chosen deliberately — its lack of learned adaptivity means its compression ratio is an unbiased probe of the latent's own entropy, and a conservative lower bound on what a learned coder could achieve on the same representation. An eight-phase curriculum then fine-tunes this model sequentially, changing exactly one aspect of the training objective per phase — spanning a float32 baseline, quantization-aware training, noise robustness, an alternative quantization proxy, a KL-regularized bottleneck that directly penalizes entropy, and increasingly detailed spectral losses — while measuring perceptual quality (PESQ-WB, STOI) and per-dimension latent entropy after every phase.

Results show entropy and quality consistently move together: spectral-loss phases raise both; the KL-regularized phase lowers both — controlled evidence that the training objective, not the quantizer, sets the latent's information content. A rate–distortion sweep across 1- to 6-bit quantization on a single fixed trained model confirms this: quality plateaus beyond approximately 4 bits despite bitrate tripling, showing the ceiling reflects the latent's information content rather than insufficient resolution. A non-causal ablation additionally rules out the real-time constraint as the source of the remaining gap to EnCodec.

References

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